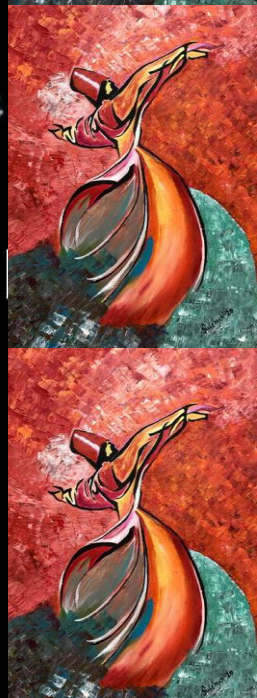




EE 354/CE 361/MATH 310 -
Introduction to Probability
and Statistics
Fall 2025



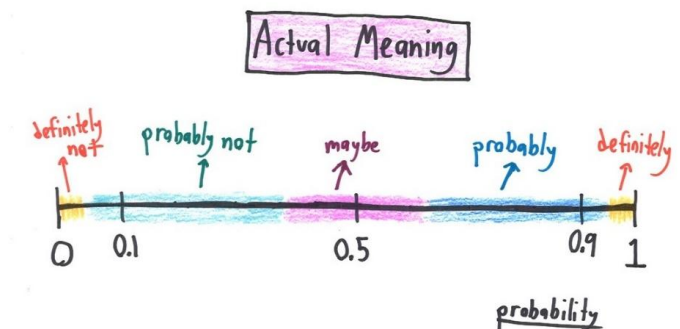


EE 354/CE 361/MATH 310 L2 section (Fall 2025)

Introduction to Probability and Statistics -

aamir hasan

Week 12 - Lecture No 22 – November 03, 2025



Announcements!

1. All lectures and grades posted.
2. Assignment No 02 uploaded – **due date November 10, 2025** (Assignment 3 to be uploaded by 2nd week of November)
 - A small correction has been made in Question No 13
3. **Quiz No 08 – November 05, 2025 (from Unit 5)**

Agenda for today

1. Unit 5 – Continuous Random Variables
 - Normal Random Variables – Gaussian Distribution

Example – Calculating Probability for a Standard Normal (Gaussian Random Variables)

$$X \sim N(0, 1)$$

1. find $P(X \geq 0)$?

$$P(X \geq 0) = 1/2$$

2. find $P(X \geq 10)$?

$$= \int_{10}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}} dx$$

No closed form available

3. How about $P(X \leq x)$?

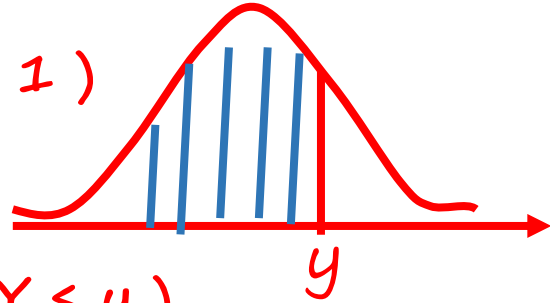
$$= \int_{-\infty}^x \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}} dx$$

No closed form CDF

Standard Normal Tables

- No closed form available for CDF
- But we have tables, for the standard normal

→ $Y = N(0, 1)$

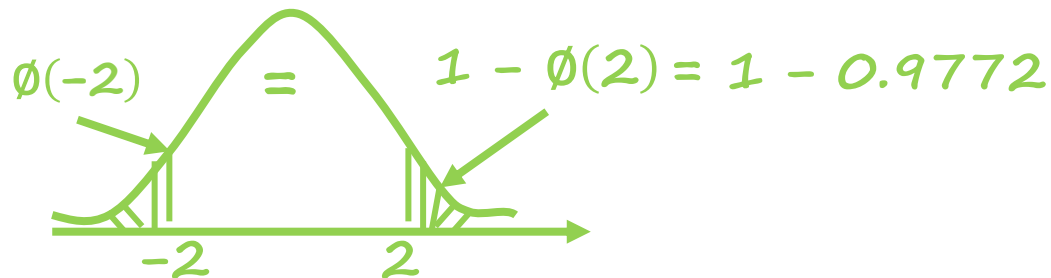


→ $\Phi(y) = F_Y(y) = P(Y \leq y)$

→ $\Phi(0) = P(Y \leq 0) = 0.5$

→ $\Phi(1.16) = 0.8770$

→ $\Phi(2.9) = 0.9981$

[illegible]

Example

Let's say $X \sim N(\mu_x, \sigma^2)$

find $P(X \geq \mu_x)$?

$$P(X \geq \mu_x) = 1/2$$

find $P(X \geq 10)$?

$$= \int_{10}^{\infty} \frac{1}{\sigma_x \sqrt{2\pi}} e^{-\frac{(x-\mu_x)^2}{2\sigma_x^2}} dx$$

No closed form available

How about $P(X \leq x)$?

$$= \int_{-\infty}^x \frac{1}{\sigma_x \sqrt{2\pi}} e^{-\frac{(x-\mu_x)^2}{2\sigma_x^2}} dx$$

No closed form CDF

What should we do?

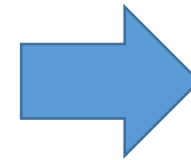
$$X \sim N(\mu_x, \sigma^2)$$

$$\begin{aligned} P(X \leq x) &= P(X - \mu_x \leq x - \mu_x) \\ &= P\left(\frac{X - \mu_x}{\sigma_x} \leq \frac{x - \mu_x}{\sigma_x}\right) \\ &= P\left(Y \leq \frac{x - \mu_x}{\sigma_x}\right) \end{aligned}$$

$$\begin{aligned} \text{Where, } Y &= \frac{X - \mu_x}{\sigma_x} \\ &= \frac{1}{\sigma_x} X - \frac{\mu_x}{\sigma_x} \end{aligned}$$

$$\begin{aligned} E[Y] &= E\left[\frac{1}{\sigma_x} X - \frac{\mu_x}{\sigma_x}\right] \\ &= \frac{1}{\sigma_x} \mu_x - \frac{\mu_x}{\sigma_x} = 0 \end{aligned}$$

$$\text{Var}(Y) = \frac{1}{\sigma_x^2} \sigma_x^2 = 1$$



$$Y \sim N(0, 1)$$

Standardizing a random variable

- Let X have mean μ and variance $\sigma^2 > 0$

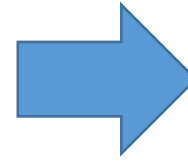
- Let $Y = \frac{X - \mu}{\sigma}$ $\Rightarrow E[Y] = 0$ $\Rightarrow \text{Var}[Y] = \frac{1}{\sigma^2} \text{Var}(X) = 1$

$$\Rightarrow X = \mu + \sigma Y$$

- Also, if X is normal, then: $Y \sim N(0, 1)$

Example (continued)

Let's say $X \sim N(\mu_x, \sigma^2)$



$$P(X \leq x_1) ?$$

$$P(Y \leq \frac{x_1 - \mu_x}{\sigma_x})$$

$$P(X \leq x) ?$$

$$= \int_{-\infty}^x \frac{1}{\sigma_x \sqrt{2\pi}} e^{-\frac{(x-\mu_x)^2}{2\sigma_x^2}} dx \Rightarrow P(Y \leq \frac{x - \mu_x}{\sigma_x})$$

$$\text{Where, } Y = \frac{X - \mu_x}{\sigma_x}$$

$$Y \sim N(0, 1)$$

Calculating normal probabilities

- Express an event of interest in terms of standard normal $P(2 \leq X \leq 8)$?

→ $X \sim (6, 4) \quad \sigma = 2$

→ $\frac{2-6}{2} \leq \frac{X-6}{2} \leq \frac{8-6}{2}$

$Y \rightarrow \text{std normal}$

$$\rightarrow P(2 \leq X \leq 8) = P(-2 \leq Y \leq 1)$$

$$= P(Y \leq 1) - P(Y \leq -2)$$

$$= P(Y \leq 1) - (1 - P(Y \leq 2))$$

$$= 0.8413 - (1 - 0.9772)$$

$$= 0.8185$$

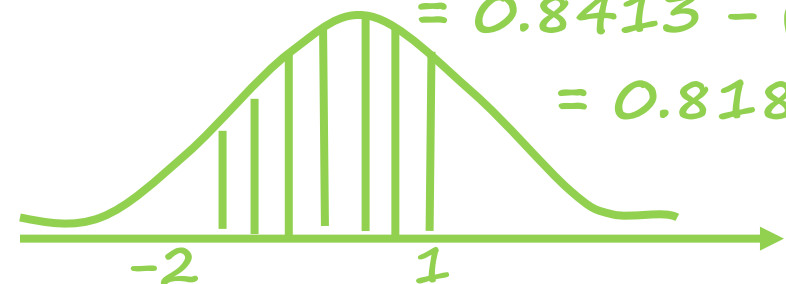


TABLE 5.1: AREA $\Phi(y)$ UNDER THE STANDARD NORMAL CURVE TO THE LEFT OF y

[illegible]

Question No 03 [25 points]

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.2	.5793	.5832	.5871	.5910	.5948	.5987	.6026	.6064	.6103	.6141
.3	.6179	.6217	.6255	.6293	.6331	.6368	.6406	.6443	.6480	.6517
.4	.6554	.6591	.6628	.6664	.6700	.6736	.6772	.6808	.6844	.6879
.5	.6915	.6950	.6985	.7019	.7054	.7088	.7123	.7157	.7190	.7224
.6	.7257	.7291	.7324	.7357	.7389	.7422	.7454	.7486	.7517	.7549
.7	.7580	.7611	.7642	.7673	.7704	.7734	.7764	.7794	.7823	.7852
.8	.7881	.7910	.7939	.7967	.7995	.8023	.8051	.8078	.8106	.8133
.9	.8159	.8186	.8212	.8238	.8264	.8289	.8315	.8340	.8365	.8389
1.0	.8413	.8438	.8461	.8485	.8508	.8531	.8554	.8577	.8599	.8621
1.1	.8643	.8665	.8686	.8708	.8729	.8749	.8770	.8790	.8810	.8830
1.2	.8849	.8869	.8888	.8907	.8925	.8944	.8962	.8980	.8997	.9015
1.3	.9032	.9049	.9066	.9082	.9099	.9115	.9131	.9147	.9162	.9177
1.4	.9192	.9207	.9222	.9236	.9251	.9265	.9279	.9292	.9306	.9319
1.5	.9332	.9345	.9357	.9370	.9382	.9394	.9406	.9418	.9429	.9441
1.6	.9452	.9463	.9474	.9484	.9495	.9505	.9515	.9525	.9535	.9545
1.7	.9554	.9564	.9573	.9582	.9591	.9599	.9608	.9616	.9625	.9633
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1.9	.9713	.9719	.9726	.9732	.9738	.9744	.9750	.9756	.9761	.9767
2.0	.9772	.9778	.9783	.9788	.9793	.9798	.9803	.9808	.9812	.9817
2.1	.9821	.9826	.9830	.9834	.9838	.9842	.9846	.9850	.9854	.9857
2.2	.9861	.9864	.9868	.9871	.9875	.9878	.9881	.9884	.9887	.9890
2.3	.9893	.9896	.9898	.9901	.9904	.9906	.9909	.9911	.9913	.9916
2.4	.9918	.9920	.9922	.9925	.9927	.9929	.9931	.9932	.9934	.9936
2.5	.9938	.9940	.9941	.9943	.9945	.9946	.9948	.9949	.9951	.9952
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2.7	.9965	.9966	.9967	.9968	.9969	.9970	.9971	.9972	.9973	.9974
2.8	.9974	.9975	.9976	.9977	.9977	.9978	.9979	.9979	.9980	.9981
2.9	.9981	.9982	.9982	.9983	.9984	.9984	.9985	.9985	.9986	.9986
3.0	.9987	.9987	.9987	.9988	.9988	.9989	.9989	.9989	.9990	.9990
3.1	.9990	.9991	.9991	.9991	.9992	.9992	.9992	.9992	.9993	.9993
3.2	.9993	.9993	.9994	.9994	.9994	.9994	.9994	.9995	.9995	.9995
3.3	.9995	.9995	.9995	.9996	.9996	.9996	.9996	.9996	.9996	.9997
3.4	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9998

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(e) [5 points] $P(X+5 < 0 \text{ and } Y+5 < 0)$

a)
$$\begin{aligned} &P(X < 2 \text{ or } X > 2) \\ &= P(X < 2) + P(X > 2) \\ &= P(-\infty \leq X < 2) + P(2 \leq X \leq \infty) \\ &= 1 \end{aligned}$$

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3.0	.9987	.9987	.9987	.9988	.9988	.9989	.9989	.9989	.9990	.9990
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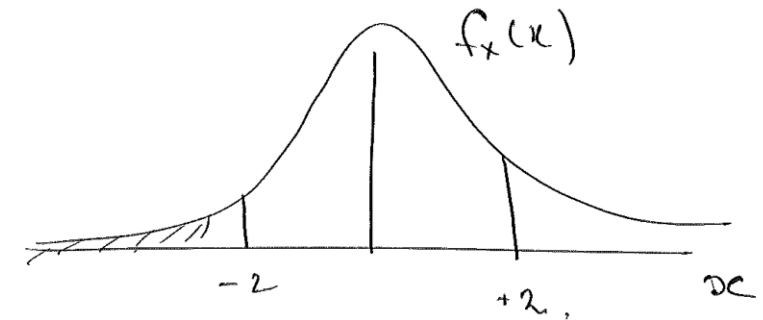
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$$\begin{aligned} (b) \quad & P(X^2 < 4) \\ &= P(-2 < X < 2) \end{aligned}$$



$$\begin{aligned} &= P(X < 2) - P(X < -2) \\ &= P(X < 2) - (1 - P(X < 2)) \\ &= 2P(X < 2) - 1 \\ &= 2(0.9772) - 1 \\ &= 0.9544 \end{aligned}$$

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2.7	.9965	.9966	.9967	.9968	.9969	.9970	.9971	.9972	.9973	.9974
2.8	.9974	.9975	.9976	.9977	.9977	.9978	.9979	.9979	.9980	.9981
2.9	.9981	.9982	.9982	.9983	.9984	.9984	.9985	.9985	.9986	.9986
3.0	.9987	.9987	.9987	.9988	.9988	.9989	.9989	.9989	.9990	.9990
3.1	.9990	.9991	.9991	.9991	.9992	.9992	.9992	.9992	.9993	.9993
3.2	.9993	.9993	.9994	.9994	.9994	.9994	.9994	.9995	.9995	.9995
3.3	.9995	.9995	.9995	.9996	.9996	.9996	.9996	.9996	.9996	.9997
3.4	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9998

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$$\begin{aligned} (c) \quad & P(X < 2 \text{ and } Y > 3) \\ &= P(X < 2) \cdot P(Y > 3) \\ &= P(X < 2) \cdot (1 - P(Y < 3)) \\ &= (0.9772) (1 - 0.9984) \\ &= 0.9772 \times 0.0016 \\ &= 0.00127036 \end{aligned}$$

Question No 03 [25 points]

X, Y are identical and independent Normal Standard RV's. Find the Probability (by showing calculations) for the following events (use the chart given on following page): -

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TABLE 5.1: AREA $\Phi(x)$ UNDER THE STANDARD NORMAL CURVE TO THE LEFT OF X

X	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
.0	.5000	.5040	.5080	.5120	.5160	.5199	.5239	.5279	.5319	.5359
.1	.5398	.5438	.5478	.5517	.5557	.5596	.5636	.5675	.5714	.5753
.2	.5793	.5832	.5871	.5910	.5948	.5987	.6026	.6064	.6103	.6141
.3	.6179	.6217	.6255	.6293	.6331	.6368	.6406	.6443	.6480	.6517
.4	.6554	.6591	.6628	.6664	.6700	.6736	.6772	.6808	.6844	.6879
.5	.6915	.6950	.6985	.7019	.7054	.7088	.7123	.7157	.7190	.7224
.6	.7257	.7291	.7324	.7357	.7389	.7422	.7454	.7486	.7517	.7549
.7	.7580	.7611	.7642	.7673	.7704	.7734	.7764	.7794	.7823	.7852
.8	.7881	.7910	.7939	.7967	.7995	.8023	.8051	.8078	.8106	.8133
.9	.8159	.8186	.8212	.8238	.8264	.8289	.8315	.8340	.8365	.8389
1.0	.8413	.8438	.8461	.8485	.8508	.8531	.8554	.8577	.8599	.8621
1.1	.8643	.8665	.8686	.8708	.8729	.8749	.8770	.8790	.8810	.8830
1.2	.8849	.8869	.8888	.8907	.8925	.8944	.8962	.8980	.8997	.9015
1.3	.9032	.9049	.9066	.9082	.9099	.9115	.9131	.9147	.9162	.9177
1.4	.9192	.9207	.9222	.9236	.9251	.9265	.9279	.9292	.9306	.9319
1.5	.9332	.9345	.9357	.9370	.9382	.9394	.9406	.9418	.9429	.9441
1.6	.9452	.9463	.9474	.9484	.9495	.9505	.9515	.9525	.9535	.9545
1.7	.9554	.9564	.9573	.9582	.9591	.9599	.9608	.9616	.9625	.9633
1.8	.9641	.9649	.9656	.9664	.9671	.9678	.9686	.9693	.9699	.9706
1.9	.9713	.9719	.9726	.9732	.9738	.9744	.9750	.9756	.9761	.9767
2.0	.9772	.9778	.9783	.9788	.9793	.9798	.9803	.9808	.9812	.9817
2.1	.9821	.9826	.9830	.9834	.9838	.9842	.9846	.9850	.9854	.9857
2.2	.9861	.9864	.9868	.9871	.9875	.9878	.9881	.9884	.9887	.9890
2.3	.9893	.9896	.9898	.9901	.9904	.9906	.9909	.9911	.9913	.9916
2.4	.9918	.9920	.9922	.9925	.9927	.9929	.9931	.9932	.9934	.9936
2.5	.9938	.9940	.9941	.9943	.9945	.9946	.9948	.9949	.9951	.9952
2.6	.9953	.9955	.9956	.9957	.9959	.9960	.9961	.9962	.9963	.9964
2.7	.9965	.9966	.9967	.9968	.9969	.9970	.9971	.9972	.9973	.9974
2.8	.9974	.9975	.9976	.9977	.9977	.9978	.9979	.9979	.9980	.9981
2.9	.9981	.9982	.9982	.9983	.9984	.9984	.9985	.9985	.9986	.9986
3.0	.9987	.9987	.9987	.9988	.9988	.9989	.9989	.9989	.9990	.9990
3.1	.9990	.9991	.9991	.9991	.9992	.9992	.9992	.9992	.9993	.9993
3.2	.9993	.9993	.9994	.9994	.9994	.9994	.9994	.9995	.9995	.9995
3.3	.9995	.9995	.9995	.9996	.9996	.9996	.9996	.9996	.9996	.9997
3.4	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9998

Question No 03 [25 points]

X, Y are identical and independent Normal Standard RV's. Find the Probability (by showing calculations) for the following events (use the chart given on following page): -

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➡ (d) [5 points] $P(XY < 0)$

(e) [5 points] $P(X+5 < 0 \text{ and } Y+5 < 0)$

$$\begin{aligned} (d) \quad P(XY < 0) &= P(X < 0) \cdot P(Y > 0) + P(X > 0) \cdot P(Y < 0) \\ &= \frac{1}{2} \times \frac{1}{2} + \frac{1}{2} \times \frac{1}{2} \\ &= \frac{1}{2} \end{aligned}$$

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3.2	.9993	.9993	.9994	.9994	.9994	.9994	.9994	.9995	.9995	.9995
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➡ (e) [5 points] $P(X+5 < 0 \text{ and } Y+5 < 0)$

$$\begin{aligned} & P(X+5 < 0 \text{ and } Y+5 < 0) \\ &= P(X < -5) \cdot P(Y < -5) \\ &= 2P(X < -5) \\ &= 2(1 - P(X < 5)) \\ &= 2(1 - 0) \\ &\approx 0. \end{aligned}$$

Conditional PDF, given an event

$$P(A) > 0$$

$$p_X(x) = \mathbf{P}(X = x)$$

$$f_X(x)\delta \approx \mathbf{P}(x \leq X \leq x + \delta)$$

$$p_{X|A}(x) = \mathbf{P}(X = x \mid A)$$

$$\Rightarrow f_{X|A}(x)\delta \approx \mathbf{P}(x \leq X \leq x + \delta \mid A)$$

$$\mathbf{P}(X \in B) = \sum_{X \in B} p_X(x)$$

$$\mathbf{P}(X \in B) = \int_B f_X(x) dx$$

$$\mathbf{P}(X \in B \mid A) = \sum_{X \in B} p_{X|A}(x)$$



$$\mathbf{P}(X \in B \mid A) = \int_B f_{X|A}(x) dx$$

Definition

$$\sum_x p_{X|A}(x) = 1$$

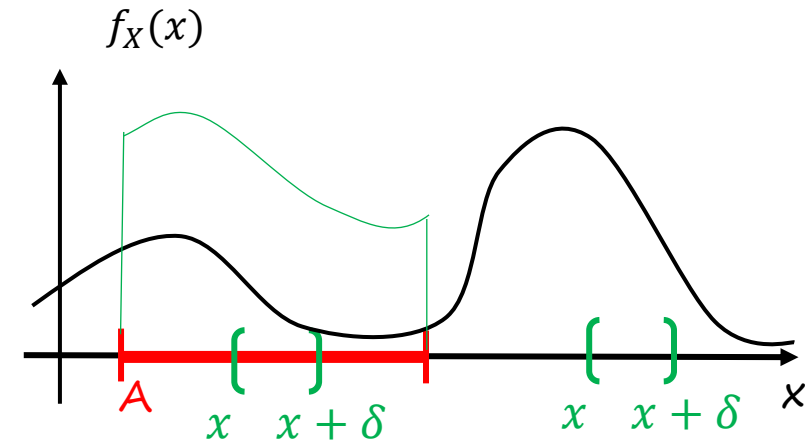
$$\int f_{X|A}(x) dx = 1$$

Conditional PDF of X , given that $X \in A$

$$\mathbf{P}(x \leq X \leq x + \delta \mid X \in A) \approx f_{X \mid X \in A}(x) \delta$$

$$\Rightarrow = \frac{P(x \leq X \leq x + \delta, X \in A)}{P(A)}$$

$$\Rightarrow = \frac{P(x \leq X \leq x + \delta)}{P(A)} \approx \frac{f_X(x) \delta}{P(A)}$$



$$f_{X \mid X \in A}(x) = \begin{cases} 0, & \text{if } x \notin A \\ \frac{f_X(x)}{P(A)}, & \text{if } x \in A \end{cases}$$

Conditional Expectation of X, given an event

$$E[X] = \sum_x x p_X(x)$$

$$E[X] = \int x f_X(x) dx$$

$$E[X | A] = \sum_x x p_{X|A}(x)$$



$$E[X | A] = \int x f_{X|A}(x) dx$$

Expected Value Rule

$$E[g(X)] = \sum_x g(x) p_X(x)$$

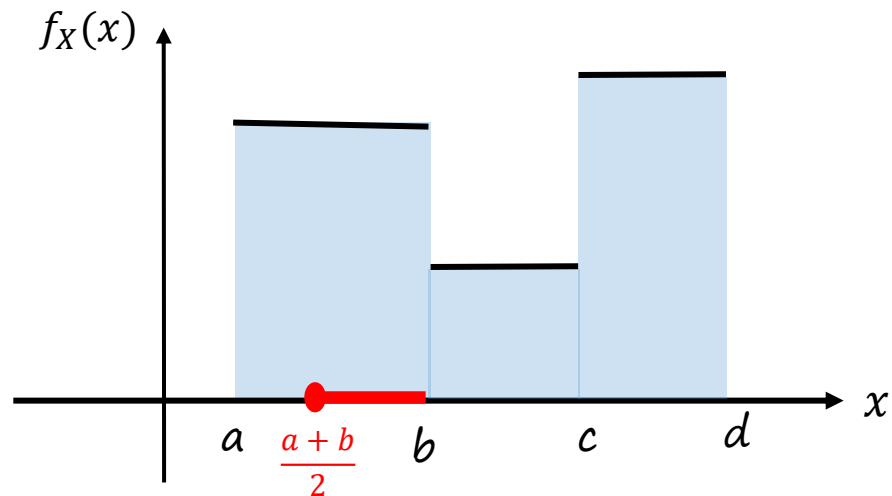
$$E[g(X)] = \int g(X) f_X(x) dx$$

$$E[g(X) | A] = \sum_x g(x) p_{X|A}(x)$$

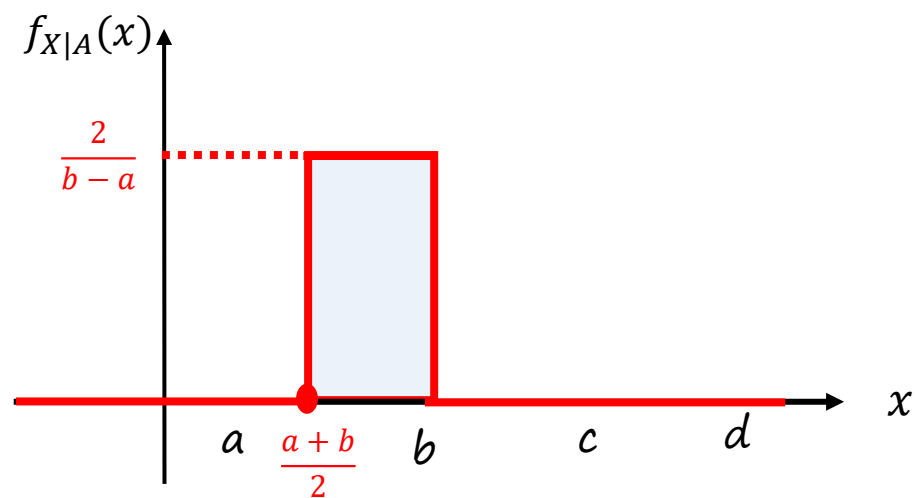


$$E[g(X) | A] = \int g(X) f_{X|A}(x) dx$$

Example



$$A: \quad \frac{a+b}{2} \leq X \leq b$$



$$\Rightarrow E[X|A] = \frac{1}{2} \frac{a+b}{2} + \frac{1}{2} b$$

$$\Rightarrow = \frac{1}{4} a + \frac{3}{4} b$$

$$\Rightarrow E[X^2|A] = \int_{\frac{a+b}{2}}^b \frac{2}{b-a} x^2 dx$$